

Meridian — Shareable native format + verified-only standard GEDCOM

Context

Meridian now has a rich model (per-person `events`, `periods` /roles, `groups` /bands, `media`; project-level `annotations` /brackets, `eras`, and `sources`). But GEDCOM export (`exportGedcom`, `src/App.jsx:75`) only writes `INDI` name/sex/BIRT/DEAT and `FAM` links, and `parseGedcom` (`src/App.jsx:112`) reads only those back — so GEDCOM is standard-compliant 5.5.1 but **lossy**, and there is no way to share a project with full fidelity between researchers.

GEDCOM is a genealogy-of-individuals format: some rich data maps to standard tags (`occu`, generic `EVEN + TYPE`, `NOTE`, `SOUR`), but project-level constructs (historical **eras**, research **brackets**) have no standard home and would only fit via non-portable custom `_` tags. Cramming everything in would produce non-idiomatic, semantically wrong GEDCOM.

The desired outcome (decisions confirmed with the user):

1. **Native full-fidelity share format** — a `.zip` bundle (`project.json` + `media/` files) that round-trips everything between Meridian users. **The JSON is the source of truth for metadata and APA7 citations**; media files in the zip are optional — if a researcher manually deletes a large scanned PDF from the zip, re-import must still retain that source's citation/metadata.
2. **Standard GEDCOM stays clean** and exports **only data explicitly marked verified**, via a new per-item `verified` flag (default: **nothing verified until marked**).
3. Non-standard-but-verified data (event-group bands; roles) is emitted as standard `NOTE` lines rather than custom tags. Eras and brackets are omitted from GEDCOM (native-only).
4. **GEDCOM import is upgraded** to read `occu`, `EVEN / TYPE / DATE` (incl. ranges), `NOTE`, and `SOUR` back into events/roles/sources; imported items land **unverified**.

Design

1. `verified` flag (data model)

- Add `verified: false` to the item factories in `src/models/project.js` (`createEvent/createPeriod/createGroup`), to `createSource`, and add optional `verified` on `birth / death` fact objects. Eras/brackets/media do not need it (never in GEDCOM).
- `normalizeProject` (`src/models/project.js`) leaves `verified` absent = falsy (unverified); no migration churn needed. Existing seeded data therefore starts unverified — intentional.

- **UI toggle:** add a "Verified" checkbox to `ItemModal` (`src/components/ItemModal.jsx`) for events/roles/groups, and to `AddSourceModal` + the source rows in `src/components/SourcesPanel.jsx` for sources. Birth/death verification: a small toggle in the focus/compare header or a dedicated person-facts editor (birth/death are edited nowhere today — add a lightweight "facts" affordance, or gate this to a follow-up and default births/deaths to the person-level via a simple modal). Simplest: reuse `ItemModal` with two synthetic "birth"/"death" fact kinds so the same verified toggle applies.
- **Visual cue:** unverified items render with a subtle dashed/reduced-emphasis treatment; verified items get a small ✓. Thread a `verified` style through `PeriodBar` / `EventDot` /group chip/ `Brace` in `src/App.jsx` and add a swatch/helper in `src/theme.js`.

2. Native .zip share format (new `src/share.js`)

- Add **JSZip** dependency (`npm i jszip`) — updates `package-lock.json` (CI already fixed to handle lockfile + npm upgrade).
- `exportProjectZip(project, storage)` : builds `project.json` (the normalized project incl. all sources metadata + APA7 citation), plus a `media/<blobKey><ext>` entry per source with a `blobKey` , loaded via `storage.loadBlob` (`src/storage/StorageService.js`). Returns a Blob for download (reuse the download pattern in `doExport` , `src/App.jsx:~515`).
- `importProjectZip(file, storage)` : unzip; parse `project.json` through `createProject` + `normalizeProject` (fresh id to avoid clobbering); for each source with a `blobKey` , if `media/<blobKey...>` exists → `storage.saveBlob` ; if the media file is absent, keep the source and its citation, set `source.fileMissing = true` and drop `blobKey` (metadata survives). Wire into `useProject` via a new `importProject(projectObj)` that saves + opens it.
- The `citation` field on sources is the canonical **APA7** string; label it "Citation (APA7)" in `SourcesPanel` . It always rides in `project.json` and (for verified sources) in GEDCOM.

3. Verified-only standard GEDCOM export (extract to `src/gedcom.js`)

Move `exportGedcom` / `parseGedcom` out of `src/App.jsx` into a testable `src/gedcom.js` , then extend `exportGedcom(people, { sources })` :

- **Always** emit the skeleton: `INDI` + `NAME` + `SEX` + `FAM` / `HUSB` / `WIFE` / `CHIL` (structure is shareable even before facts are verified).
- **Gate facts on `verified`** : `BIRT` / `DEAT` only if `birth.verified` / `death.verified` ; events, roles, groups, and source citations only when the item's `verified` is true.
- **Mapping:** events → `1 EVEN <label> + 2 TYPE <type> + 2 DATE <year> (+ 2 NOTE )` ; roles/periods → `1 OCCU <label> + 2 DATE FROM <start> TO <end>` ; groups/bands → `1 NOTE <label> (start-end)` ; a linked verified source → a `0 @Sn@ SOUR` record (`1 TITL` , `1 REPO` , APA7 as `1 TEXT` / `1 NOTE`) plus `2 SOUR @Sn@` citation under the item. Eras/brackets omitted.
- A toast summarizes what was exported (e.g. "Exported 12 verified facts across 5 people").

4. Upgraded GEDCOM import (`src/gedcom.js`)

Extend `parseGedcom` to walk deeper: `OCCU` → a `period` (parse `DATE FROM..TO`), `EVEN` with `TYPE / DATE / NOTE` → an `event` , `NOTE` under `INDI` → a `group` or note, `SOUR` records + 2 `SOUR @ref@` → `sources` + `sourceId` links. All imported items are created via the factories so they get ids and `verified: false` . Keep the birth-year filter behavior.

5. Export/Import UI (`src/App.jsx`)

Replace the two toolbar buttons with an **Export** ▼ (→ "Meridian project (.zip — full)" / "GEDCOM (.ged — verified only)") and **Import** that accepts both `.zip` and `.ged` , dispatching by extension to `importProjectZip` or `parseGedcom` . Reuse `Modal` / `btnStyle` .

Critical files

- `src/models/project.js` — `verified` in factories; APA7 note on `createSource` .
- `src/gedcom.js` — **new**: extracted + extended export/import (verified filtering, EVEN/OCCU/NOTE/SOUR).
- `src/share.js` — **new**: JSZip export/import with citation-survives-missing-media tolerance.
- `src/components/ItemModal.jsx` — Verified toggle (+ optional birth/death fact kinds).
- `src/components/SourcesPanel.jsx` — Verified toggle + "Citation (APA7)" label.
- `src/hooks/useProject.js` — `importProject(projectObj)` (save + open a new project).
- `src/App.jsx` — Export/Import menu, wiring, verified visual cues on lane items.
- `src/theme.js` — verified/unverified style helper.

Dependencies

- `jszip` (new runtime dep). Touches `package-lock.json` ; CI already handles lockfile sync + npm upgrade.

Verification

1. **Unit tests** (`src/tests/gedcom.test.js` , `src/tests/share.test.js` , extend `models.test.js`):
 - `verified` defaults false on all factories.
 - GEDCOM export: unverified project → skeleton only (INDI/FAM, no BIRT/DEAT/EVEN); after marking facts verified → they appear with correct standard tags; output parses back via `parseGedcom` .
 - GEDCOM import: a fixture with `OCCU` , `EVEN` + `TYPE` , `NOTE` , `SOUR` → yields periods/events/ sources, all `verified:false` .
 - Share: `exportProjectZip` → `importProjectZip` round-trips people/eras/annotations/sources; then simulate a zip with the media file removed → source + APA7 citation retained, `fileMissing:true` .
2. **Build + suite**: `npm run build` , `npm test` (58 existing + new all green).

3. **Manual** (`npm run dev` , <http://localhost:5173/meridian/>): mark a few items verified; Export GEDCOM and confirm it's valid 5.5.1 containing only verified facts (+ NOTE for groups); Export the `.zip` , unzip, delete a media file, re-import → citation survives and item shows file-missing; import a real third-party `.ged` and confirm names/dates/occupations/notes load.
4. Reload to confirm imported/verified state persists through IndexedDB autosave.